People's Democratic Republic of Algeria

2022-2023 Ministry of Higher Education and Scientific Research

National Higher School of Mathematics

NATIONAL HIGHER SCHOOL OF MATHEMATICS

المدرسة الوطنية العليا في الرياضيات

Final exam (2^h30)

Name : Groupe:

Exercice 1. Among these assertions, specify which ones are true and which ones are false.

(0.25×20)=5pts

	True	False
1-The median is always higher than the mean.		$\surd$
2- If $P(A\cap B)\geq 0.10$ then $P(A)\geq 0.10$.	$\surd$	
3- If $P(A\|B)=0.5$ and $P(B)=0.5$, then A and B are necessarily independent.		$\surd$
4- If a value is higher than the mean then it is an outlier.		$\surd$
5- $P(A)-P(B)\neq P(A\cap B^c)$.		$\surd$
6- Multiplying all values by 2, will multiply the STD by 2.	$\surd$	
7- Removing an outlier that is negative, will most likely decrease the STD.	$\surd$	
8- If A and B are mutually exclusive events, then $P(A\|B)=P(A)$.		$\surd$
9- The mean is equal to 0 for a symmetric histogram.		$\surd$
10- $E[X+2Y]=E[X]+2E[Y]$.	$\surd$	
11- The probability is always non negative.	$\surd$	
12- The STD of a sample is always less than 1.		$\surd$
13- Two r.v X and Y are identically distributed if $\exists x\in\mathbb{R}, F_X(x)=F_Y(x)$.		$\surd$
14- The median is always higher than the mean.		$\surd$
15- If $A\cap B\neq\emptyset$ and the events B and C are disjoint. then $A\cap B\cap C^c=\emptyset$		$\surd$
16- If $P(A\|B)=0.6$ and $P(B)=0.2$ then $P(A\cap B)=0.12$.	$\surd$	
17- $P(A_1\cup A_2\cup ...)\leq P(A_1)+P(A_2)+...$	$\surd$	
18- The Coefficient of determination R^2 is a unitless number between -1 and 1.		$\surd$
19- $\forall a,b,c\in\mathbb{R}, Var(aX+bY+c)=a^2Var(X)+b^2Var(Y)+2abCov(X;Y)$.	$\surd$	
20- We can learn the IQR by looking at the boxplot.	$\surd$	

Exercice 2. For a sample of 15 students we observe the Time (in minutes, X) usually spent using Facebook per day, and the final grade in the Statistics exam (Y):

x_i	0	11	17	16	22	17	25	30	27	27	31	35	30	45	60
y_i	25	22	28	30	22	27	26	21	27	28	23	29	30	24	27

1. Compute mean, median and quartiles for both X and Y

$$\overline{X}=\frac{1}{15}\sum_{i=1}^{15}x_i=\frac{393}{15}=26.2 \quad\text{and}\quad \overline{Y}=\frac{1}{15}\sum_{i=1}^{15}y_i=\frac{389}{15}=25.933. \qquad (0.5)$$

There are 15 (odd) observation, the medians $\widetilde{X}$ and $\widetilde{Y}$ are the center of observation in the ordered lists. The location of the medians are the $\left(\frac{15+1}{2}\right)^{th}$ item :

$$\widetilde{X}=x_{(8)}=27 \quad\text{and}\quad \widetilde{Y}=y_{(8)}=27. \qquad (0.5)$$

Also, Q_1 and Q_3, are the $\left(\frac{15+1}{4}\right)^{th}$ and $\left(\frac{3\times(15+1)}{4}\right)^{th}$ items in the ordered lists :

$$Q_1^X=x_{(4)}=17, Q_3^X=x_{(12)}=31 \quad\text{and}\quad Q_1^Y=y_{(4)}=23, Q_3^Y=y_{(12)}=28. \qquad (1)$$

2. Find the standard deviation and the coefficient of variation for both X and Y. Which variable is more scattered?

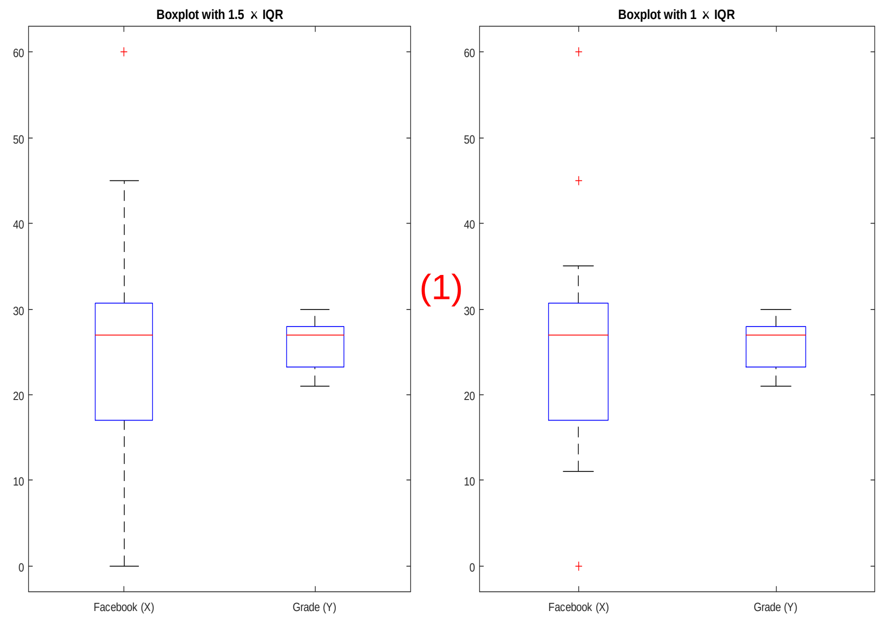
$$s_X = \sqrt{\frac{1}{15-1}\sum_{i=1}^{15}\left(x_i - \overline{X}\right)^2} = \sqrt{\frac{2836.4}{14}} = 14.234 \qquad (0.5)$$

and

$$s_Y = \sqrt{\frac{1}{15-1}\sum_{i=1}^{15}\left(y_i - \overline{Y}\right)^2} = \sqrt{\frac{122.93}{14}} = 2.963 \qquad (0.5)$$

$CV_X = \frac{s_X}{\overline{X}} \times 100\% = 54.328\%$ and $CV_Y = \frac{s_Y}{\overline{Y}} \times 100\% = 11.426\%$. (0.25)
Thus the time spent on Facebook (X) is more heterogeneous than the Statistics grade (Y). (0.25)

3. Plot the box-plot for both X and Y and comment them.

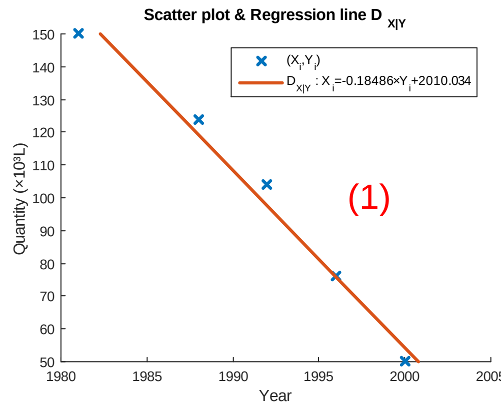


(0.5)
Both boxplots show an asymmetrical character with presence of outliers for the time spent on Facebook (X)

Exercice 3. Since clean-air regulations have dictated the use of unleaded gasoline, the supply of leaded gas has diminished. The table below was given

Year (X)	1981	1988	1992	1996	2000
Quantity $(\times 10^3 L)$ (Y)	150	124	104	76	50

1. Draw the scatter plot between the two variables and measure the correlation between them.



$$\overline{X} = 1991.4, \overline{Y} = 100.8,$$

$$Cov(X,Y) = \frac{1}{5}\sum\left(x_i - \overline{X}\right)\left(y_i - \overline{Y}\right) = -227.92$$

$$Var(X) = \frac{1}{5}\sum\left(x_i - \overline{X}\right)^2 = 43.04$$

$$\sigma_X = \sqrt{43.04} = 6.56$$

$$Var(Y) = \frac{1}{5}\sum\left(y_i - \overline{Y}\right)^2 = 1232.96(\times 10^3 L)^2$$

$$\sigma_Y = \sqrt{1232.9} = 35.113(\times 10^3 L)$$

$$\rho(X,Y) = \frac{Cov(X,Y)}{\sigma_X \sigma_Y} = \frac{-227.92}{6.56 \times 35.113} = -0.989\,49 \qquad (1)$$

2. Create a linear model to predict the quantity that would be available in 2005.

To predict the quantity that would be available in 2005, we need first to fit the line

$$Y = aX + b + \varepsilon.$$

therefore, $\widehat{a} = \frac{Cov(X,Y)}{Var(X)} = \frac{-227.92}{43.04} = -5.2955$ and $\widehat{b} = \overline{Y} - \widehat{a}\overline{X} = 100.8 - (-5.2955) \times 1991.4 = 10646.26$. Thus

$$Y = -5.2955X + 10646.26 + \varepsilon$$

and

$$Y_{2005} = -5.2955 \times 2005 + 10646.26 = 28.783(\times 10^3 L)$$ (1)

3. Create a model that would estimate the year when leaded gasoline will first become unavailable then draw this line on the above scatter plot.

The year when leaded gasoline will first become unavailable is the year with $Y = 0$ in the model

$$X = a'Y + b' + \varepsilon'.$$

with $\widehat{a'} = \frac{Cov(X,Y)}{Var(Y)} = \frac{-227.92}{1232.96} = -0.18486$ and $\widehat{b'} = \overline{X} - \widehat{a'}\overline{Y} = 1991.4 - (-0.18486) \times 100.8 = 2010.034$. Thus

$$X_0 = -0.18486 \times (0) + 2010.034 \simeq 2010.$$ (1)

4. Give the value of R^2 for the line of the 3^{rd} question then briefly interpret its value.

$$R^2 = \frac{V_E(X)}{Var(X)} = \frac{Var(X)\rho(X,Y)^2}{Var(X)} = (-0.98949)^2 = 0.97909$$ (1)

The coefficient of determination R^2 summarizes the part of the variance explained by the regression lines. It is a unitless number, between 0 and 1.The regression lines here explain 97.91% of the variance (of X and Y) which gives that the goodness of fit is very good.

Exercice 4. Consider a given population, a certain disease D and a certain symptom S. We know that 85% of the population members who have the disease D do have the symptom S, while the remaining 15% do not. Suppose that 95% of those who do not have the disease D do not have the symptom S, while 5% do have the symptom S (due to some other cause).
Suppose that 10% of the given population have the disease.

1. What is the probability that a random person chosen from the population has the symptom?

We have

$$P(D) = 0.1, P(D^c) = 0.9$$
$$P(S|D) = 0.85, P(S^c|D) = 0.15$$
$$P(S^c|D^c) = 0.95, P(S|D^c) = 0.05$$

and we're looking for $P(S)$ which can be obtained by the **Law of Total Probability**

$$P(S) = P(S|D)P(D) + P(S|D^c)P(D^c) = 0.85 \times 0.1 + 0.05 \times 0.9 = 0.13$$ (2)

2. Given that a random person was sampled, and he has the symptom, what is the probability that he or she has disease?

Since we already know that he has the symptom, the probability that he or she has disease is given by **Bayes' Theorem**

$$P(D|S) = \frac{P(D \cap S)}{P(S)} = \frac{P(S|D)P(D)}{P(S)} = \frac{0.85 \times 0.1}{0.13} = 0.65 \qquad (2)$$

3. What is the probability that a random person does not have the disease and does not have the symptom?

$$P(D^c \cap S^c) = P(S^c|D^c)P(D^c) = 0.95 \times 0.9 = 0.855 \qquad (1)$$